# PHILOSOPHY OF MATHEMATICS

## 16–1 WHAT IS PHILOSOPHY?

The word "philosophy" is a combination of two Greek words and means literally *love of wisdom*. In the early Greek civilization, philosophy was the study of all branches of knowledge. As man's learning increased through the ages, certain disciplines grew until they split away from philosophy and became separate areas for study. We no longer think of medicine, law, mathematics, physics, chemistry, biology, economics, etc., as parts of philosophy, although philosophy was the father of all these sciences. That is, the first men who called attention to important concepts in mathematics, physics, chemistry, economics, political science, etc., were philosophers. It was philosophy that suggested how knowledge about these new areas should be collected and organized.

Philosophy has always concerned itself with the unknown—with those situations in which man is not sure of himself. Because of this, many of the conclusions that philosophers reach cannot be checked by experiment, as we check physical laws, and consequently many people have the mistaken belief that time devoted to philosophical thought is largely wasted.

It is impossible to describe briefly and clearly just what the subject matter of philosophy consists of today. Some of the branches into which philosophy is subdivided will be listed here, but we make no pretense of completeness.

*Moral philosophy*, or *ethics*, investigates the nature of right and wrong. It is concerned with distinguishing between good and bad conduct. It raises questions like, "Is an action to be judged good or bad according to its effect or its intent?" "What are *good* and *evil*?" "How shall we determine whether one action is better or worse than another?"

*Religious philosophy* is the study of concepts common to religions. On the other hand, *theology* is concerned with the logical structure of a single religion.

*Political philosophy* does not consider details of government, but rather the principles upon which various systems of government are constructed. "How much authority may the state exert over an individual?" "Are there fundamental human rights, which no state can usurp? If so, what are they?"

*Metaphysics* ("beyond" physics) is sometimes described as the study of final or ultimate meanings. Much that is nonsense has been written in this area, but it includes many deep and fascinating insights.

*Epistemology* can be described briefly as the theory of knowledge. It considers the general questions, "How do we learn?" and, "How shall we organize our knowledge?" As a matter of fact, the philosophy of *mathematics* is just a branch of epistemology; so you may think of this chapter as an introduction to a fragment of epistemology, which is itself a fragment of philosophy.

It is impossible to discuss the philosophy of mathematics without considering the philosophy of science, for mathematics is used as a tool in every science to organize and interpret scientific knowledge. We shall mention how mathematics thus often leads to the discovery of new scientific facts.

## 16–2 THE TASK OF SCIENCE

All information about what we call the *real* or *external* world is obtained from the five senses: sight, sound, taste, smell, and touch. Our sense organs, for example the eye, send messages to the brain. A newborn baby cannot do much with these messages, but as he grows older, he learns to interpret them. We are always amazed at the accomplishments of a person who is both blind and deaf (for example, Helen Keller), for it is difficult for us to grasp how such a person can have much understanding of the world. What we are inclined to forget is that all of us have learned how to interpret information from our sense organs so as to make useful responses. As you read this page, your eye receives impressions and sends messages to the brain. You have learned to interpret and respond to these messages.

*Every scientific observation is, at first, a sense impression, that is a touch, smell, taste, sight, or sound, for some experimenter. It is the task of scientists to understand, explain, and predict happenings in the world of our sense impressions.*

Some philosophers have suggested that because our only knowledge of the external world comes from sense perception, *we cannot even prove that the external world has any existence apart from our sense perception of it; that perhaps the world that we believe we see, touch, and smell exists only as things exist in a dream.*

The story is told that the great 17th-century philosopher Bishop Berkeley was so convinced by this line of thought that one day, when he was walking and meditating upon philosophical ideas, he encountered a large stone in his path and so much doubted the actual existence of the stone that he drew back his foot and gave it a vigorous kick. We have no record of the good bishop's thoughts as he hobbled home.

## 16–3 KINDS OF SCIENCE

The world is very complicated, and we have little understanding of it. To simplify the task of understanding our environment, certain topics that seem somewhat related are studied by themselves as separate sciences. For example, biology is the study of living organisms, and biology is itself separated into several subsciences, like zoology, the study of animal life, and botany, the study of plant life. To understand all aspects of biology, it is necessary to know some chemistry and physics. For this reason chemistry and physics are regarded as more basic, or more fundamental, than biology. Of the sciences concerned with the real world, the most fundamental is physics, which deals with the properties of matter.

All sciences become *quantitative* when they have progressed far enough. This means that measurements are made and numbers are introduced into the sciences. Those things which we measure, and to which we assign numbers, are called *quantities*. As soon as numbers are introduced into a science, mathematics becomes necessary for its study. As a matter of fact, one needs geometry in order to discuss the physics of motion of bodies, even before one makes measurements. Even psychology and sociology, two of the newer sciences that have only recently split away from philosophy, need statistics in order to organize and interpret the huge number of facts with which they deal. Thus all the sciences depend upon mathematics, so that it has been termed "the handmaiden of the sciences."*

Sometimes mathematics itself is classified as one of the sciences; and sometimes one hears the phrase "science *and* math-

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*E. T. BELL, *The Handmaiden of the Sciences*. Baltimore: The Williams and Wilkins Co., Baltimore, 1937.

ematics." In any event, call it what you may, mathematics is different from the other sciences in that it is *independent of experiments*. *Mathematics is a creation of the mind alone.*

## 16–4 IS PHYSICS TRUE?

We cannot give an acceptable definition of physics in a few words, but it will not take long to make the question of this section meaningful. You have studied a little physics previously, so you should know that physics is concerned with forces, motion, heat, sound, electricity, and much else. For example, in that part of physics called mechanics, it is learned that if at some instant we know the position and velocity of a ball, and if in addition we know the forces acting upon it, then we can *calculate* where it will be at any later time.

You need not concern yourself here with the exact method used to make this calculation. What is important is that *given certain information, we can predict what will happen*. Physics, and indeed every science, is just as "true" as its predictions are *reliable*. As soon as a physical theory fails to predict what will happen, then it becomes necessary to construct a new theory that will predict more accurately. In physics, the final test of "truth" is whether the predictions we get from our theory actually agree with our observations. As an example of this, Aristotle had written in his *Physics* that heavier bodies would fall more rapidly than lighter ones. This prediction is a logical consequence of Aristotle's physical *theories*, or *laws*. Every school child knows how Galileo *proved* these theories *false* by dropping balls of various sizes and weights from the leaning tower of Pisa. Every observer who was open-minded enough to watch the experiment (and there were indeed some who were not) became convinced that Aristotle's laws of motion were false and that it was necessary to replace them by new laws that would come closer to predicting the events that were actually observed.

The physical laws used by the physicist to make his predictions are actually *assumptions* and are expressed in terms of mathematical formulas. If the mathematical calculations employed do not lead to correct predictions, this does not mean that the mathematics is incorrect.* Instead it means that the physical laws do not agree well enough with reality, that the *physical assumptions* are at fault.

When the physicist defines physical concepts like force, weight, etc., and then gives physical laws that tell how these concepts are related, he is creating what we call a *mathematical model*. What he is saying is that *if* there are things in the real world that correspond to these concepts, and *if* these things are related to one another according to the rules called physical laws, *then* mathematics tells us what will happen in the real world.

If the predictions come out wrong, then the physicist must construct a better mathematical model. Often it happens that the physicist cannot solve the mathematical problems that arise within the mathematical model. Then he calls upon the mathematician for help. In this way physics supplies mathematics with interesting problems.

As an example of a simple mathematical model, consider our solar system. The physicist thinks of the sun and planets as points that are tracing out curves as they move through space. By using calculus and analytic geometry, these curves are described by mathematical equations that predict how the planets ought to move through the sky if the law of gravitation is valid.

It is most interesting that early in the 18th century, when astronomers calculated how the planets *should* move according to the law of gravitation, it was noticed that one of the planets,

*Of course someone could make a computational mistake, but this is the fault of the computer, not of the mathematics he is using.

Uranus, did not follow the exact path predicted for it by the mathematical model. The astronomers had so much faith in the mathematical model that they became sure that there must be an "invisible" planet, which was pulling Uranus off its course with its gravitational attraction. The Englishman John C. Adams and the Frenchman LeVerrier set about attempting to solve the problem of determining the path of the "invisible" planet.* Working with the mathematical model, they computed where it *ought* to be to affect the course of Uranus as it did. When observers focused their telescopes upon that part of the heavens, they saw the planet Neptune precisely where the mathematical model predicted that it would be found. This is but one of the many dramatic illustrations of how an *abstract* mathematical model can lead us to the discovery of new facts about our universe.

## 16–5 IS MATHEMATICS TRUE?

If by this question we mean, "Does the real world behave like the mathematical models that we construct?" then we are asking a nonmathematical question. The *truth* of mathematics is independent of the real world. The *usefulness* of one kind of mathematics or another does depend upon the real world.

*Pure mathematics is concerned with the logical consequences of postulate systems.* In pure mathematics, postulate systems are studied primarily because they are interesting, and often without regard to any future use. In the history of mathematics, there are many postulate systems that have been studied for their own sakes and only much later found to be applicable to the external world. For example, the non-Euclidean geometry needed in the theory of relativity was first studied many years before any physical application of it was made.

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*A fascinating account of this work can be found in J. R. Newman's *The World of Mathematics*. Vol. 2, pp. 822–839.

*Applied mathematics is concerned with those parts of pure mathematics that are of greatest use in the science of today.* Scientific applications often suggest the constructions of new postulate systems. Mathematics and science thus serve as stimuli to each other.

We have not yet given a direct answer to the question, "Is mathematics true?" What is "truth," anyway? Each of us has some sort of feeling about what "truth" means. Truth, for the physicist, is *that which works*, and this attitude is characteristic of science. Truth, in practical affairs, usually refers to *whether some event actually took place*. Here, we often take the word of others whom we trust. Truth, in spiritual matters, depends upon *faith*. In general, agreement among people regarding spiritual beliefs is difficult to obtain, for there is no test by experience that we can all agree upon.

When we say that a system of mathematical postulates is "true," we mean only, *these postulates are free from contradiction*. That is, if it is impossible to prove from our postulates that a statement is true, and then also prove that the same statement is false, in this case and only in this case do we call our postulates true. Remember that we do not assert in mathematics that anything actually exists in the physical world. Rather, our theorems have the form: *If* a set of things has so-and-so properties, *then* it has such-and-such properties.

We have developed our geometry this year from statements that dealt with a set of points that we called our plane. A mathematician is naturally pleased to encounter in the physical world a set of objects that actually has the properties he has postulated for one of his mathematical systems, for then he can be sure that the set he finds will have other properties that he has established by studying his abstract system. A natural question to ask at the close of this year's study of geometry is whether there actually exists in the real world a set of things that satisfies the postulates of our geometry. Does the space we live in actually have the properties of the Euclidean space we study in geometry?

The answer will probably not satisfy you. The answer is, *we do not know. Furthermore, no mathematician ever expects to know.* Our geometric plane consists of *infinitely* many points. *It will never be possible to determine whether our universe is finite or infinite in extent.* Although we have no assurance that any part of the real world actually fits our geometric postulates, experience has shown us that much of our geometric theory is applicable to the world around us; and we can indeed use our geometry to make very accurate predictions.

## 16–6 FREEDOM FROM CONTRADICTION

In this section we discuss a very deep question. When we have a postulate system before us, how do we know that these postulates are *consistent*? How do we know that we shall never deduce from these postulates, by correct logical reasoning, two contradictory conclusions? Let us give an example, first given by Bertrand Russell,* to illustrate what we mean.

*Russell's barber.* Let us assume that, in a certain small town, there is but one barber. We assume that this barber shaves everyone in the town who does not shave himself, and no one else.

We now ask who shaves the barber.

*If* the barber does not shave himself, *then* he is a person who does not shave himself, and therefore he is shaved by the barber—himself. Since this is impossible, we have proved that the statement *"The barber does not shave himself"* is false. Hence, the barber shaves himself. But, by our assumptions, *if* the barber shaves himself, *then* he is one of the men who is not shaved by the barber—again himself. This proves that the statement *"The*

*An English philosopher who is interested in the foundations of mathematics. At about the turn of the century, he wrote *Principia Mathematica* with A. N. Whitehead. In the book they attempt to derive all mathematics from logic alone.

*barber shaves himself"* is false, and hence, *"The barber does not shave himself,"* is true. But now we have arrived at a contradiction, having shown from our postulates that the statement *"The barber does not shave himself" is both true and false.*

This example shows that in forming a set of postulates, we must be careful that our postulates are consistent; that is, that they do not lead to contradictory conclusions. When our postulate sets are very simple, it is sometimes possible to prove that no contradiction can ever be reached by logical reasoning. For more complicated postulate sets, this may not be possible.

David Hilbert spent much of the latter portion of his life trying to prove the consistency of certain interesting mathematical systems. At about the same time that he wrote his *Foundations of Geometry*, he showed that geometry is consistent if arithmetic is consistent. He accomplished this proof by using the ideas of Chapter 15. The details need not concern us here. All we need know is that our geometry is consistent if arithmetic is. We should remark that non-Euclidean geometries have also been proved to be consistent, providing that arithmetic is. Of course this has nothing to do with the question of whether the space we live in is Euclidean or non-Euclidean.

Now, how do we know that arithmetic is consistent? The answer may again seem disappointing, for the answer is, *we do not know!* Perhaps we never will know! There is no proof that we can never be led to an impossible conclusion in arithmetic. For example, we do *not* know that it is impossible to *prove*, by a valid chain of logical arguments, that $0 = 1$!

All mathematicians believe that arithmetic is consistent. This belief is based upon faith, not proof. Of course, arithmetic is useful. It works, and helps us reach conclusions that turn out to be correct. But here is the position in which the scientist finds himself. If an inconsistency is discovered in arithmetic, the postulates for arithmetic will have to be changed. The possibility seems to be remote, so it does not worry working mathematicians too much.

## 16–7 MATHEMATICAL RESEARCH

Euclid's geometry is more than 2000 years old. The average person holds the erroneous opinion that all of mathematics was discovered long ago, and that today we study only old stuff. There are, however, many journals published throughout the world that are devoted primarily to *new* mathematics. In fact, during the past fifty years, more new mathematics has been invented* than in all previous history! Mathematical progress is stimulated by both intellectual curiosity and the needs of modern science. Because of the mathematical nature of modern science and engineering, mathematicians are in greater demand than ever before and command excellent salaries.

It is impossible to describe here the breadth and depth of current mathematical research. As yet, we have not undertaken enough elementary mathematics to appreciate such a description. However, in your study of plane geometry, you have made a fine beginning by studying a single mathematical system in some detail. The logical features of geometry are common to all mathematics, so that you have before you one example of a postulate system and some of its consequences.

## 16–8 MATHEMATICS AND LIFE

How may an understanding of the philosophy of mathematics affect you as an individual in your everyday life? Mathematicians are much like any other group. If you were to attend the International Congress of Mathematicians, which convenes

*Is mathematics *discovered* or *invented*? The first word suggests that mathematics is "sitting around waiting to be found." The second suggests that mathematics is created by the mind, that human activity is essential.

once every four years, you would see a group of people differing widely in dress, speech, religious and political beliefs, and physical appearance. But you would, if you were observant enough, see many common characteristics. Everyone there would have a way to communicate with everyone else—not through English, French, German, or Russian, but through the universal symbolism of mathematics. You would soon realize that these folk, while actively interested in the real world, were still far more interested in the world of ideas.

You would see that each one present had a tremendous respect for the personalities and intellectual capabilities of his neighbors.

The philosophy of mathematics speaks to those who will listen and assures them that even more important than wresting food, shelter, and material riches from the earth is the acquisition of knowledge and understanding.

## SUGGESTED READING

From *The World of Mathematics*, by J. R. Newman.

1. Philip E. B. Jourdain, "The Nature of Mathematics." Vol. 1, pp. 4–71.
2. J. W. N. Sullivan, "Mathematics as an Art." Vol. 3, pp. 2015–2021.
3. Raymond L. Wilder, "The Axiomatic Method." Vol. 3, pp. 1647–1667.